Scenario 4: Satellite Orbit Collision

👋 **Hello, Hacker!** Today's mission: take one satellite you already control and slam it into another one in a nearby orbit. No special break-in, just an ordinary orbit-adjustment command with the right values. Take the operator's seat and let's go.

Mission map (the flow)

> **1 INTEL → 2 PROPAGATE & TIME → 3 SOLVE → 4 TRANSMIT → RESULT**

The step indicator at the top shows which stage you are on. Clear each stage's "To do" top to bottom, then move on.

Your mission

As the operator, send ENIGMA-1 an ordinary orbit-adjustment command that makes it collide with AURORA-2 on the next orbit. The result plays out on the side screen (monitor 2).

Mission briefing

ENIGMA-1 is a satellite you already control. This is not a break-in: you change the values of a legitimate propulsion (delta-v) command to push the satellite into a neighbour's orbit, and the collision debris then threatens other members of the same constellation. On the start screen, read the background, pick a difficulty (Easy/Normal/Hard), and begin. The difficulty sets how precise the burn must be to count as a collision.

STAGE 1: Read the orbits (INTEL)

On screen: Left is the intercepted TLE text, right is the decoded table of six orbital elements.

To do: Press PARSE TLE to decode both satellites' TLEs (ENIGMA-1 and AURORA-2) into the six orbital elements (a, e, i, Ω, ω, M). Check the values, then CONFIRM INTEL to move on.

STAGE 2: Pick the impact moment (PROPAGATE & TIME)

To do: Use the time slider to run the clock and move the satellites forward and back. Find the moment AURORA-2 reaches the point where the two orbits cross, and lock that as the impact time. If unsure, turn on the hint buttons to reveal the crossing point and the valid times. Once the time is set, CONFIRM TIME.

STAGE 3: Solve the burn (SOLVE)

To do (in order):

1. **Assemble the equation chain:** Click the cards in dependency order to complete the calculation flow. You must finish the chain before you can transmit.
2. **Tune the burn values:** Raise the orbit with Δv prograde (along the direction of motion), fine-tune where the orbits meet with Δv cross-track (perpendicular to the orbit plane), and match the arrival time with Δv phasing.
3. **Pick a thruster:** Choose the thruster for the burn.
4. **Check:** Turn on the hint button to see the geometry (MOID) and timing gauges for how close you are, then CONFIRM BURN.

STAGE 4: Transmit the command (TRANSMIT)

To do: Your solved values are assembled into a real command packet (CCSDS/cFS). Press TRANSMIT BURN COMMAND to send it to ENIGMA-1 and watch the satellite move on the side screen (monitor 2). If it does not collide, monitor 2 shows a near miss, so press RESET & RE-PLAN and adjust your values.

MISSION CLEAR: Collision and debris cascade

Mission clear. A single legitimate orbit-adjustment command turned a satellite into a weapon. On monitor 2 you saw the collision debris spread to other members of the same constellation. One satellite's problem cascading into many is exactly the danger we have to prevent in real orbits.